

MATH 3260 - Chapter 4 Worksheets

Vector Spaces

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 4.1 - Vectors in $\mathbb{R}^n$

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Let $\mathbf{u} = (5, -2)$ and $\mathbf{v} = (-3, 7)$. Find $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - \mathbf{v}$, and $-\frac{1}{2}\mathbf{v}$.
2. Let $\mathbf{a} = (2, 1, -1)$, $\mathbf{b} = (0, 3, 2)$, and $\mathbf{c} = (1, -2, 4)$. Compute $3\mathbf{a} - 2\mathbf{b} + \mathbf{c}$.
3. Solve $2(\mathbf{x} - \mathbf{u}) = \mathbf{v} + \mathbf{x}$ for $\mathbf{x}$ when $\mathbf{u} = (1, 2, 3)$ and $\mathbf{v} = (4, -1, 2)$. First isolate $\mathbf{x}$ algebraically, then substitute components.
4. Determine coefficients c_1, c_2 such that
$$(9, 21) = c_1(1, 1) + c_2(1, -1).$$
5. A badminton player makes court-movement vectors $(2, 1)$, $(-1, 3)$, and $(0, -2)$ meters. Find the total displacement and interpret the two components.

Section 4.2 - Vector Spaces

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For each set with the usual operations, determine whether it is a vector space. Justify with the relevant axioms or by identifying one axiom that fails.
 - (a) all (x, y, z) satisfying $x + y - z = 0$;
 - (b) all (x, y) satisfying $x + y = 1$;
 - (c) all polynomials of degree at most three;
 - (d) all 2×2 matrices with determinant 1;
 - (e) all continuous functions f satisfying $f(0) = 0$.
2. Explain why testing only several numerical examples cannot prove that a set satisfies the vector-space axioms for every possible vector and scalar.

Section 4.3 - Subspaces of Vector Spaces

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

For every subspace test, explicitly check: (1) the zero vector of the parent space, (2) closure under addition, and (3) closure under scalar multiplication.

1. Determine whether

$$W = \{(x, y, z) : 2x - y + 3z = 0\}$$

is a subspace of $\mathbb{R}^3$.

2. Determine whether

$$W = \{(x, y, z) : 2x - y + 3z = 7\}$$

is a subspace of $\mathbb{R}^3$.

3. Determine whether

$$W = \{p \in P_3 : p(0) = p(1) = 0\}$$

is a subspace of P_3 .

4. Determine whether the set of symmetric 2×2 matrices is a subspace of $M_{2,2}$.

5. Determine whether the set of invertible 2×2 matrices is a subspace of $M_{2,2}$.

6. Find a parametric description and a spanning set for

$$\{(x, y, z) : x + y - z = 0\} \subseteq \mathbb{R}^3.$$

Section 4.4 - Spanning Sets and Linear Independence

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Determine whether $(8, 3, 2)$ is a linear combination of

$$(1, 2, 0), \quad (0, 1, 1), \quad (2, 0, 1).$$

If so, find the coefficients.

2. Determine whether

$$\{(1, 0, 1), (0, 1, 1), (1, 1, 2)\}$$

is linearly independent.

3. Does $\{(1, 2), (3, 6)\}$ span $\mathbb{R}^2$? Explain both algebraically and geometrically.

4. Determine whether

$$\{1 + x, x + x^2, 1 + 2x + x^2\}$$

is independent in P_2 .

5. Find a nontrivial dependence relation among

$$(1, 0, 2), \quad (0, 1, 1), \quad (2, 1, 5).$$

6. Explain geometrically why two nonparallel vectors in $\mathbb{R}^2$ span the plane.

Section 4.5 - Basis and Dimension

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Determine whether

$$\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$$

is a basis for $\mathbb{R}^3$. Check both spanning and independence.

2. Find the dimension of

$$\text{span}\{(1, 2, 0), (0, 1, 1), (1, 3, 1)\}.$$

3. Give a standard basis and the dimension of P_4 .

4. Give a basis and the dimension of the vector space of symmetric 2×2 matrices.

5. Find a basis and the dimension of

$$W = \{(s + t, s - t, 3s) : s, t \in \mathbb{R}\} \subseteq \mathbb{R}^3.$$

6. Explain why five vectors in P_3 must be linearly dependent. Relate your answer to $\dim(P_3)$.

Section 4.6 - Rank, Nullspace, and Systems

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation. For

$$B = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ -1 & -2 & 1 & 0 \end{bmatrix},$$

complete the following.

1. Find the RREF, state the pivot columns using one-based numbering, and find $\text{rank}(B)$.
2. Find a basis for the row space.
3. Find a basis for the column space using columns of the *original* matrix.
4. Find a basis for $N(B)$ and the nullity.
5. Verify

$$\text{rank}(B) + \text{nullity}(B) = 4.$$

6. If $\mathbf{x}_p = (1, 0, 2, -1)^T$, compute $\mathbf{b} = B\mathbf{x}_p$ and write the complete solution of $B\mathbf{x} = \mathbf{b}$ as

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h, \quad \mathbf{x}_h \in N(B).$$

Section 4.7 - Coordinates and Change of Basis

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

Coordinates depend on an **ordered** basis. In this worksheet, parentheses indicate the order explicitly.

1. For the ordered basis

$$B = ((1, 2), (2, -1)),$$

find $[(8, 7)]_B$.

2. Reconstruct $\mathbf{x}$ from

$$[\mathbf{x}]_B = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

using the same ordered basis B .

3. Find the coordinates of $p(x) = 3 - 2x + 4x^2$ relative to the ordered basis $(1, x, x^2)$.

4. Find the coordinates of the same polynomial relative to the ordered basis

$$(1, x - 1, (x - 1)^2).$$

5. Let

$$B = ((1, 1), (1, -1)), \quad S = ((1, 0), (0, 1)).$$

Find $P_{S \leftarrow B}$ and $P_{B \leftarrow S}$.

6. Use the transition matrix to convert

$$[\mathbf{x}]_B = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

to standard coordinates. State in one sentence what the transition matrix accomplishes.

Section 4.8 - Applications of Vector Spaces

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Verify that $\cos x$ and $\sin x$ solve $y'' + y = 0$.
2. Compute their Wronskian and use it to justify linear independence. Explain what independence means for the general solution of the differential equation.

3. Compute the Wronskian of 1, x , and x^2 .

4. Identify the center and semiaxes of

$$\frac{(x-3)^2}{16} + \frac{(y+1)^2}{9} = 1.$$

5. Identify the vertex and opening direction of

$$(y-2)^2 = -12(x+1).$$

6. Use the 45° change of coordinates to remove the cross term in

$$4x^2 + 4xy + 4y^2 = 12.$$

Classify the resulting conic and explain what the coordinate change accomplished.